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Applied Theoretical Physics
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Project 1 for '*Computational Physics - Material Science*', SoSe 2025

Project Title: The Electrostatic Signatures of Water Nanodroplets

Goal: Water nanodroplets play a significant role in atmospheric sciences (clouds, flashes, etc.) and sorption and transport of ionic material in nanocomposites as well as in chemical catalysis. Objective of this project is to set up an MD simulation of a water nanodroplet and analyze its structural and electrostatic properties.

The simplest water model consists of three interaction sites, with the geometry of a water molecule being defined by three internal degrees of freedom: two O–H bond lengths, r_{OH_1} and r_{OH_2}, and one H–O–H angle, θ_{HOH} (see Fig. 1).

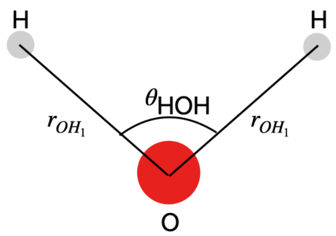

Abbildung 1: Geometry of a water molecule.

The general functional form of the interaction potential, V, can be written as the sum of the intramolecular contributions, V_{intra}, and the intermolecular contributions, V_{inter}. Assuming simple harmonic potentials for the intramolecular interactions, V_{intra} is defined as:

$$V_{intra} = \frac{k_{bond}}{2}[(r_{OH_1} - r^0_{OH})^2 + (r_{OH_2} - r^0_{OH})^2] + \frac{k_{angle}}{2}(\theta_{HOH} - \theta^0_{HOH})^2 \tag{1}$$

where r^0_{OH} and θ^0_{HOH} represent the equilibrium bond length and angle, respectively. k_{bond} and k_{angle} are the bond and angle strength, respectively. V_{inter} is defined as:

$$V_{inter}(r_{ij}) = \left(V_{LJ}(r_{ij}) + V_{q_iq_j}(r_{ij})\right)$$

where $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ is the distance between atoms i and j, $V_{LJ}(r_{ij})$ is a LJ potential for atom pair (i,j), and $V_{q_iq_j}(r_{ij})$ is short-range spherically truncated, charge-neutralized, damped, and shifted potential to treat the Coulombic (electrostatic) interaction. Let us note that, we only assume LJ interaction between oxygen atom, i.e. $\epsilon_{OO} \neq 0$ while $\epsilon_{OH} = \epsilon_{HH} = 0$, while Coulomb interactions are only calculated between atom from two different molecules. $V_{q_iq_j}(r_{ij})$ between the atom i with charge q_i and position $\mathbf{r}_i$ and the atom j with charge q_j and position $\mathbf{r}_j$ separated by a distance $r_{ij} < R_c$ is given by

$$V_{q_iq_j}(r_{ij}) = \frac{q_iq_j}{4\pi\epsilon_0}\left[\frac{\mathrm{erfc}(\alpha r_{ij})}{r_{ij}} - \frac{\mathrm{erfc}(\alpha R_c)}{R_c} + \left(\frac{\mathrm{erfc}(\alpha R_c)}{R_c^2} + \frac{2\alpha}{\sqrt{\pi}}\frac{\exp\left(-\alpha^2 R_c^2\right)}{R_c}\right)(r_{ij} - R_c)\right] \tag{2}$$

where α and R_c are the damping parameter and the cutoff distance, respectively, and erfc() is the complementary error-function.

The objective of the project is to simulate a water droplet with MD in the NVT ensemble at finite temperature, and to quantify its structure and electrostatic potential. The project is build up in successive tasks. Students are encouraged to follow the time schedule given in Fig 2.

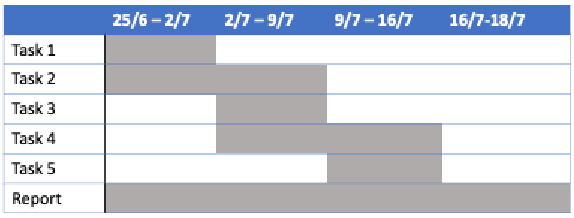

Abbildung 2: Workplan for a successful completion of the project.

Task 1. Angular potential

- Derive analytically the force, $\mathbf{F}_{angle}$, resulting from the angle contribution (see Eq. 1) that needs to be implemented.
- Implement the angular and the bond contribution in a MD engine with $R^0_{OH} = 0.1$ nm, $\theta^0_{HOH} = 104.5°$, $k_{bond} = 1058$ kCal/mole/Å^2, $k_{angle} = 75$ kCal/mole/rad^2, $m_H = 1.6735 \times 10^{-27}$ kg, and $m_O = 2.6561 \times 10^{-26}$ kg.
- Simulate the trajectory of a single water molecule over 1000 time steps with a timestep of $\Delta t = 0.5$ femtoseconds at a temperature of $T_d = 300$ K. Record and plot the time evolution of the H–O–H angle $\theta(t)$ and the mean O–H bond length $\langle r_{\mathrm{OH}} \rangle(t)$, and compare their oscillation frequencies with those predicted by the corresponding harmonic oscillator models using appropriate reduced masses. Report the mean bond lengths $\langle r_{\mathrm{OH}_1} \rangle$, $\langle r_{\mathrm{OH}_2} \rangle$, and the mean bond angle $\langle \theta_{\mathrm{HOH}} \rangle$. Conclude on the validity and numerical stability of the implementation based on the consistency of the results with expected physical behavior.

Task 2. Implementation of the MD code for a water nanodroplet

Implement a MD code in the NVT ensemble with the following functionalities

- bond potential,
- angular potential,
- LJ interactions between O and O using ϵ_{OO} and σ_{OO},
- Coulomb interaction using Eq. 2,
- a Berendsen thermostat,
- no periodic boundary conditions (in that sense $V = \infty$, while of course the droplet itself has a finite volume. Due to the strong water-water attraction hardly any water molecules will leave the droplet during the simulation time but it can happen. If it happens unexpectedly too frequently, PBCs have to be used).

Task 3. Set up the MD for the water nanodroplet

Simulate the trajectory of a system containing $M = 216$ H_2O molecules initially distributed on a cubic lattice of a virtual simulation box of volume $V = 216\sigma^3_{OO}$ for $50000\Delta t$ with $\Delta t = 0.5$ fs (the first 10000 steps being used to equilibrate the temperature at $T_d = 300$K using a Berendsen thermostat with $\tau = 500\Delta t$). The equilibrium bond length and angle are the ones defined in Task 1. The LJ parameters are $\epsilon_{OO} = 0.2 k_B T_d$ and $\sigma_{OO} = 3.21$ $\mathring{A}$, and the partial charges are $q_O = -0.84e$ and $q_H = 0.42e$. The cutoff radius for both the LJ and the Coulomb interactions is $R_c = 2.5\sigma_{OO}$, and the damping parameter, α for the coulomb interaction is $\alpha = 1./R_c$. Provide an Ovito animation with oxygen and hydrogen colored in red and white, respectively, to illustrate the formation of the nanodroplet. To validate the droplet formation and moving into equilibrium, calculate the mean kinetic and potential energies. Can you calculate a droplet radius versus time? Can you calculate a second moment of the mass distribution with respect to the center-of-mass? Are there other quantities to check that you arrived at a meaningful equilibrium?

Task 4. Analysis

After equilibration, in the production runs,

a) Calculate and plot the radial water (one-body) number density profile, $\rho(r)$, with r being the distance of the oxygen atoms to the droplet center-of-mass, averaged over the whole production run. What do you observe?

b) How does it compare with the (experimental) number density of bulk water at conditions of $T = 300$ K and 1 bar pressure?

c) Compare to the hydrogen number density profile in the same plot. Can you say something about the structure of water at the interface?

d) The water molecules have a permanent dipole which is a vector. How are these vectors on average oriented with respect to the radial vector from the droplet center to the water molecules' centers-of-mass?

Task 5. Real world context - be creative and play around with these optional suggestions

a) Calculate the radial electrostatic field and the electrostatic potential (i.e. as a function of r, the radial distance from the droplet center-of-mass) of the nanodroplet by numerically integrating the charge density (equals the number density profiles weighted by the partial charges) using the Poisson equation. The potential can be interpreted as a voltage. Is it negative or positive? Compare to 9V from a normal battery and the voltage of a typical lightning flash.

b) Find LJ parameters of sodium (+1 charge) and Iodine (-1 charge) ions in literature and include a pair of those (Na^+I^-) into the droplet. What do they do? Do they pair in the center or on the surface? Or do they even separate because one ion is 'surface-active'? Note that surface-active ions are important for catalysis and interface charge release and exchange. How much work is required to remove a single ion from the droplet?

c) Interpret and discuss your results, preferably to real-world contexts. For your talk, do you find examples in reality where charged nanodroplets are important, e.g., in chemical applications, or in the atmospheric sciences? What is actually the latest knowledge about how lightning flash is generated in the atmosphere?

Numerical values of quantities to be used:

quantity	value (units)
k_B	1.38×10^{-23} (m^2 kg s^{-2} K^{-1}
T	300 (K)
m_H	1.6735×10^{-27} (kg)
m_O	2.6561×10^{-26} (kg)
Δt	0.5×10^{-15} (s)
τ	$500\Delta t$
t_{eq}	$10000\ \Delta t$
t_{prod}	$40000\ \Delta t$